

Clinical Assessment of Familial Intellectual and Learning Impairment with Reduced Head Circumference

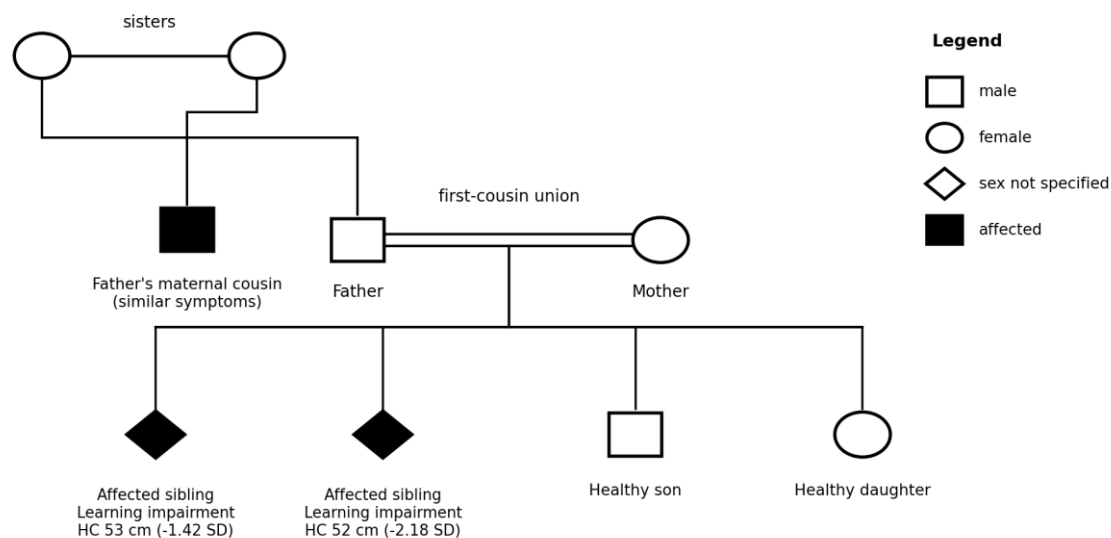
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1. Clinical background

The supplied case description reports a consanguineous couple (first cousins; described as daughter of a paternal aunt and son of a maternal uncle) with two affected adult offspring showing learning impairment. At examination, the affected individuals were 32 and 35 years old, with head circumferences of 53 cm (-1.42 SD) and 52 cm (-2.18 SD), respectively. The family also includes one healthy son and one healthy daughter. The father's maternal cousin is also reported to show similar symptoms.

2. Pedigree



Pedigree reconstructed only from the supplied case description; sex of the two affected siblings was not stated.

Figure 1. Pedigree reconstructed from the supplied case description.

The parents are reported as first cousins (daughter of a paternal aunt and son of a maternal uncle). Two offspring are affected with learning impairment and reduced head circumference, while one son and one daughter are healthy. The father's maternal cousin is also reported to have similar symptoms. The sex of the two affected siblings is not stated in the source, so neutral pedigree diamonds are used for them.

3. Analysis strategy

1. Starting dataset: The annotated ANNOVAR spreadsheet contains approximately 570,550 variants.

2. Filter 1 - Func.refGene: The Func.refGene column was filtered to retain exonic variants, reducing the dataset to approximately 24,214 variants.
3. Filter 2 - CLINSIG: The ClinVar clinical-significance column (CLINSIG) was filtered to retain entries containing Pathogenic, including combined classifications containing Pathogenic.
4. Filtered result: The filtering workflow reduced the dataset to 19 variants for subsequent review.
5. Interpretation: The filtered variants were reviewed in the context of the family phenotype and the expected inheritance pattern.

4. Variants retained after filtering

The filtering workflow retained variants in the following 19 genes:

#	Gene
1	ALG6
2	F5
3	FMO3
4	CYP1B1
5	TTN
6	COG5
7	CFTR
8	RPL11
9	IMPAD1
10	GRHPR
11	ABCC2
12	HBB
13	MYO7A
14	ROBO3
15	AIPL1
16	TCAP
17	DYM
18	ABHD12
19	PRODH

5. Candidate assessment: TBC1D32

Gene	TBC1D32
Transcript (as read from the spreadsheet)	NM_152730
Exon	14
Coding change	c.1466A>T

Protein change	p.E489V
Variant class	Missense / nonsynonymous SNV
Genotype in analyzed sample	Homozygous alternate (1/1)
SIFT	D (deleterious prediction; screenshot showed score approximately 0.02)
Population filter	Passed preliminary 1000 Genomes frequency filter (<1%)
Current status	Candidate variant; pathogenicity not established

TBC1D32 is biologically and clinically relevant because current ClinGen curation classifies the TBC1D32-ciliopathy gene-disease relationship as Definitive with autosomal-recessive inheritance (ClinGen, 18 April 2025). The ClinGen evidence summary notes roles in ciliary function and Sonic Hedgehog signaling and includes reported individuals with microcephaly and developmental phenotypes. One reported 10-year-old female had intellectual disability, mild facial dysmorphism and microcephaly with a homozygous missense TBC1D32 variant. This supports prioritizing TBC1D32 as a gene, but it does not prove that the family's p.E489V variant is pathogenic.

6. Recommended validation and segregation work

The next step is to validate the candidate and test whether it follows the disease pattern in the family. Sanger sequencing can be used both to confirm the NGS/WES call and to genotype available relatives.

Family member	Expected genotype if simple AR model is correct	Interpretation
Affected sibling 1	1/1	Supports co-segregation
Affected sibling 2	1/1	Supports co-segregation
Father (unaffected)	0/1	Expected carrier
Mother (unaffected)	0/1	Expected carrier
Healthy son	0/0 or 0/1	Should not be 1/1 under a fully penetrant simple AR model
Healthy daughter	0/0 or 0/1	Should not be 1/1 under a fully penetrant simple AR model
Affected paternal-side relative	Ideally 1/1 if the same familial cause is responsible	Especially informative, but the exact pedigree relationship must be confirmed

If both affected siblings are homozygous, both parents are heterozygous carriers, and unaffected siblings are not homozygous, the evidence for co-segregation becomes substantially stronger. If the second affected sibling does not carry the candidate in the expected state, TBC1D32 becomes a weaker explanation and other candidates/models should be revisited.

7. Practical Sanger workflow

1. Confirm the genomic coordinates and transcript/HGVS notation for c.1466A>T (p.E489V).
2. Design primers flanking the variant in exon 14, avoiding common polymorphisms and repetitive sequence.
3. PCR-amplify the target region from the proband and available relatives.

4. Perform bidirectional Sanger sequencing.
5. Inspect chromatograms at the candidate position and assign genotypes (0/0, 0/1, 1/1).
6. Compare genotype with phenotype across the pedigree (segregation analysis).
7. Integrate segregation with population frequency, computational evidence, phenotype match and established gene-disease validity before applying ACMG/AMP classification.

8. References

1. Family case description supplied for the project.
2. ClinGen. TBC1D32-ciliopathy gene-disease validity curation. Classification: Definitive; autosomal recessive. Evaluation date: 18 April 2025.
3. Harris SC et al. 2023. PMID: 36826837.